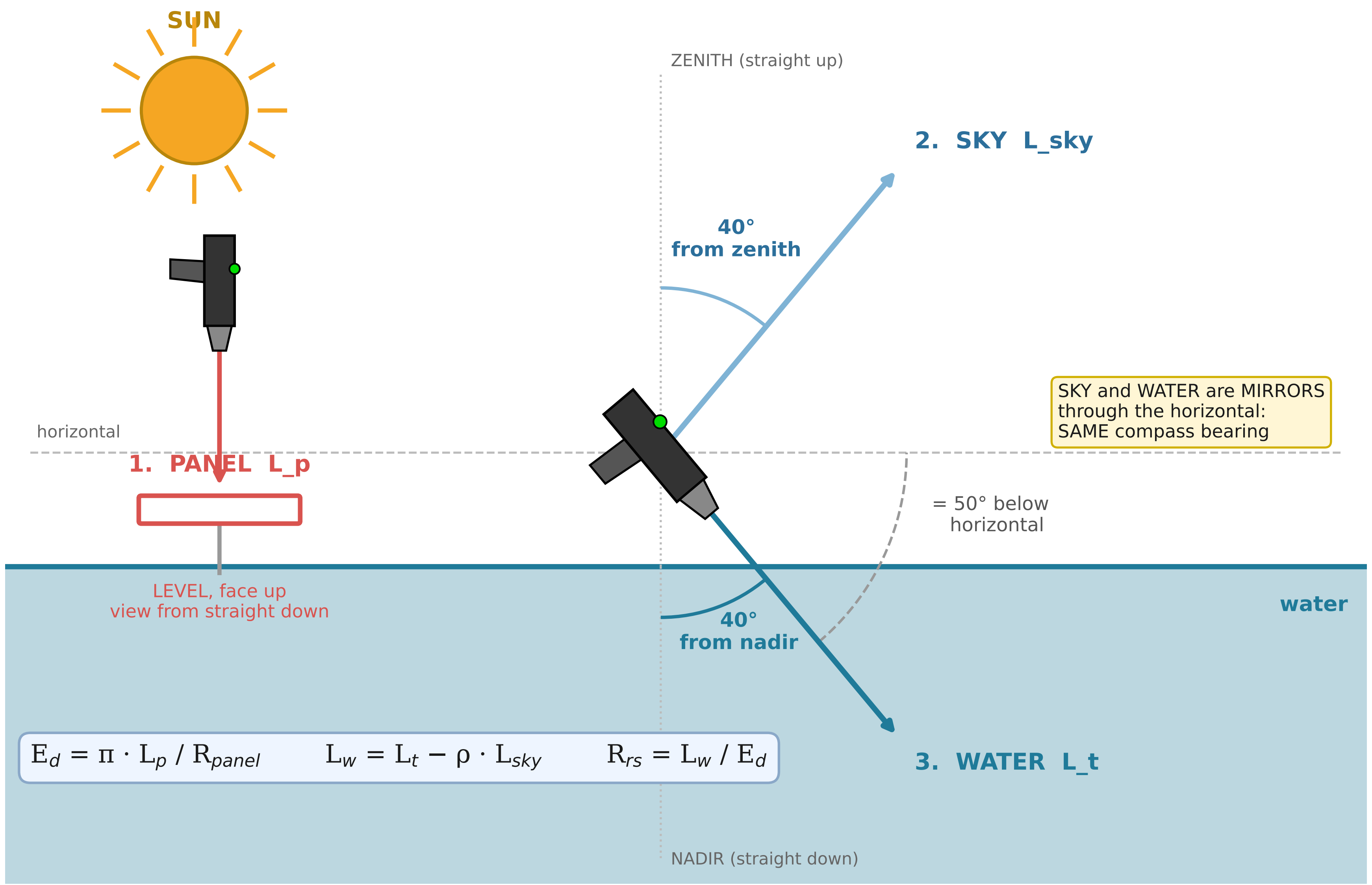
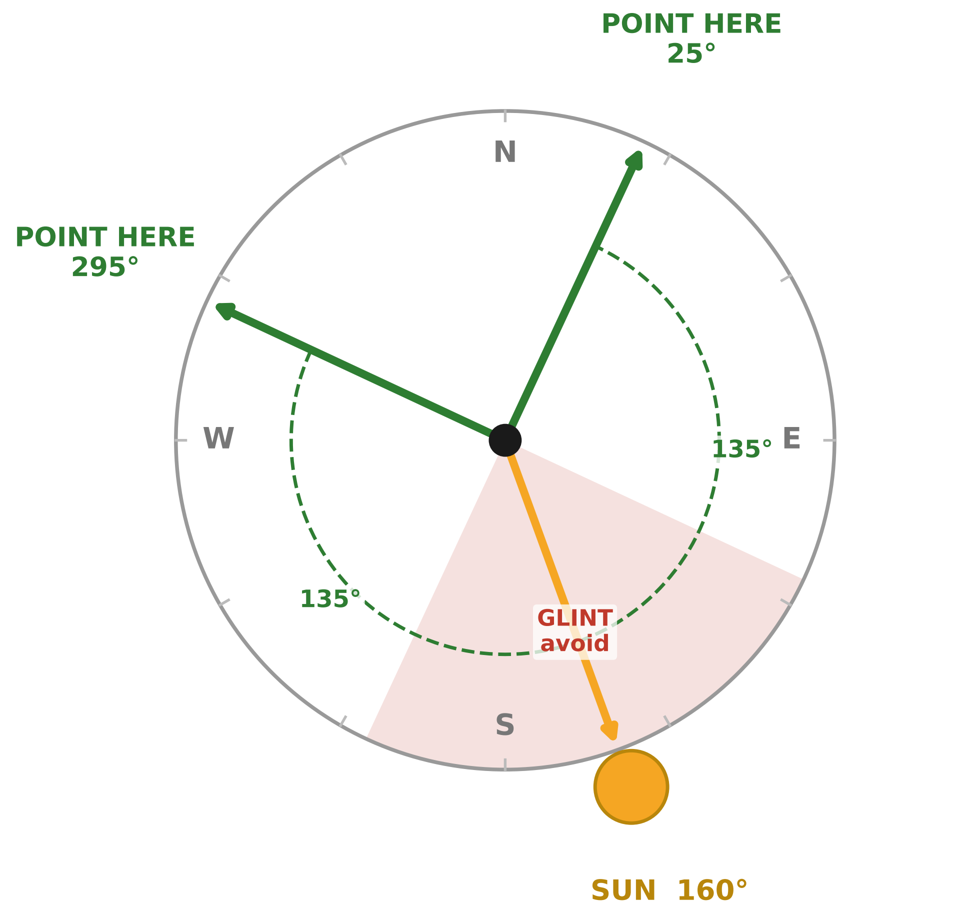
ABOVE-WATER R_{rs} — three-scan field method · Spectral Evolution NaturaSpec Plus

Geometry after Mobley (1999): view 40° from nadir, 135° in azimuth from the sun. Sky scan is the MIRROR of the water scan.

A. SIDE VIEW — the three scans and their angles



B. PLAN VIEW — which way to point (looking down)



Both bearings are 135° in azimuth from the sun. Pick whichever is clear of the boat, its shadow and its wake. The GUI computes these for you from latitude, longitude, date and time.

1. PANEL

White reference panel (Spectralon),
LEVEL and face up, in full sun.
No shadow on it, yours included.
View from straight down.
Do not touch the white face.

DARWin: take as the REFERENCE.
Re-take every ~10 min, and the
moment the light changes.

2. SKY

Sky at 40° from ZENITH
= 50° ABOVE horizontal.

SAME compass bearing as the
water scan. Not the opposite
one. This is the single most
common field mistake.

DARWin: TARGET scan. Save.

3. WATER

Water at 40° from NADIR
= 50° BELOW horizontal,
135° in azimuth from the sun.
FROM A BOAT OR PIER use 90°:
135° looks back at the hull
or its shadow (IOCCG v3.0).
No glint, foam or wake.

DARWin: TARGET scan. Save.
Repeat 2 and 3 five times.

RECORD

WIND SPEED (m/s) — sets ρ, the
largest error in the whole
measurement. ρ = 0.028 only
valid below ~5 m/s.
Wind DIRECTION + a photo
Cloud, or sun obscured?
Water clear or turbid?
turbid → do NOT use nir_zero
Panel reflectance used (0.99?)

The instrument already logs
Range, Tilt, GPS Time and
Solar Angle (= ELEVATION,
not zenith — verified).

CHECK ON THE SPOT

⚠ THE CLOCK MAY BE WRONG.
Use GPS Time for solar
geometry. On a real file the
clock read 05:56 with the
sun below the horizon.

R_rs positive across 400-700?
negative = over-subtracted
Peak where the water looks?
Replicates on top of each other?
R_rs(443) order 1e-3 to 1e-2?

FOV × range = footprint. An 8°
lens at 3.8 m sees ~0.5 m of
water; the GUI plots it.

Copy the .sed files off tonight.